Week 0 – Git, GitHub and project basics

Goal: Understand how to organise your project and save your work.

No.	Question	Answer in very simple words	Example / command
1	What is a project?	A project is a collection of files that work together to solve a problem.	Online Examination System
2	What is a folder?	A folder stores files and other folders.	backend, frontend
3	What is VS Code?	A code editor used to write and manage project files.	Open your Exam folder in VS Code.
4	What is a terminal?	A place where we type commands to control the computer.	VS Code → Terminal → New Terminal
5	What is Git?	Git is a tool that tracks changes in your project.	git --version
6	What is GitHub?	GitHub is an online platform for storing and sharing Git repositories.	Your online project repository
7	What is a repository?	A repository is a project folder tracked by Git.	git init
8	What is git init?	Starts Git tracking in the current folder.	git init
9	What is git status?	Shows which files have changed.	git status
10	What is git add?	Selects changed files to be included in the next commit.	git add .

11	What is a commit?	A saved snapshot of your project changes.	git commit -m "Add exam feature"
12	What is a branch?	A separate line of work where you can develop changes.	git branch
13	What is git push?	Uploads your local commits to GitHub.	git push
14	What is git pull?	Downloads and merges changes from the remote repository.	git pull
15	What is a Git tag?	A label attached to a particular commit, often used for versions.	git tag v1.0
16	What is .gitignore?	A file that tells Git which files not to track.	Ignore .venv and .env.

Week 2 – Web application fundamentals, REST APIs and authentication

Goal: Understand how your frontend, backend, database and authentication work together before implementing them.

A. Web application fundamentals

No.	Question	Answer in very simple words	Online Examination System example
1	What is a frontend?	The part of an application that users see and interact with.	React login page and exam dashboard.
2	What is a backend?	The part that processes requests and performs application logic.	FastAPI checks exam submissions.
3	What is a database?	A place to store application information.	PostgreSQL stores exams and results.
4	What is a three-tier architecture?	An application divided into presentation, application and data layers.	React → FastAPI → PostgreSQL.
5	What is the presentation layer?	The user interface.	React frontend.
6	What is the application layer?	The business logic and processing layer.	FastAPI backend.
7	What is the data layer?	The layer that stores and retrieves information.	PostgreSQL.
8	What is HTTP?	A protocol used for communication between web clients and servers.	React sends an HTTP request to FastAPI.

9	What is HTTPS?	HTTP protected using TLS encryption.	Secure communication with a deployed website.
10	What is SSL/TLS?	Security technology used to encrypt network communication.	Protects login and API traffic.
11	What is a client?	The application that sends a request.	React browser application.
12	What is a server?	The application that receives and processes a request.	FastAPI server.
13	What is an API?	A way for two applications to communicate.	React communicates with FastAPI.
14	What is JSON?	A text format for sending structured data.	Exam details sent to React.
15	What is a request?	A message sent to a server asking it to do something.	Request to list exams.
16	What is a response?	The server's reply to a request.	List of exams returned as JSON.

B. HTTP methods, REST and CRUD

No.	Question	Answer in very simple words	Example
17	What is REST?	A common style for designing web APIs around resources.	Exam API.
18	What is a resource?	An item or thing managed by an API.	Exam, question, result.
19	What is GET?	Reads information.	GET /api/exams
20	What is POST?	Creates a new item or submits data.	POST /api/exams
21	What is PUT?	Replaces an existing item.	PUT /api/exams/1
22	What is PATCH?	Updates part of an existing item.	PATCH /api/exams/1
23	What is DELETE?	Removes an item.	DELETE /api/exams/1
24	What is CRUD?	Create, Read, Update and Delete.	Manage examination records.
25	What is a path parameter?	A value included in the URL path.	/api/exams/5 – exam ID is 5.
26	What is a query parameter?	An optional value added after ? in a URL.	/api/exams?status=active
27	What is a request body?	Data sent with a request, usually JSON.	New exam details.
28	What is a status code?	A number explaining the result of an HTTP request.	200, 201, 404.

C. HTTP status codes

Status	Meaning	Simple example
200	OK	Exams were returned successfully.
201	Created	A new exam was created.
204	No Content	An exam was deleted successfully.
400	Bad Request	The request is not acceptable.
401	Unauthorized	Missing or invalid login token.
403	Forbidden	Student tries an admin-only action.
404	Not Found	Exam ID does not exist.
422	Validation Error	Required field is missing or invalid.
500	Internal Server Error	Unexpected backend problem.

D. Authentication and authorization

No.	Question	Answer in very simple words	Example
29	What is authentication?	Checking who you are.	Student logs in.
30	What is authorization?	Checking what you are allowed to do.	Student cannot create exams.
31	What is the difference between authentication and authorization?	Authentication checks identity; authorization checks permission.	Login first, then check admin role.
32	What is a session?	Server-side information that remembers a user's login.	A login session maintained by a server.
33	What is a JWT?	A signed token that carries claims about a user.	Keycloak access token.
34	What is a claim?	A piece of information inside a token.	Username, issuer or role.
35	What is RBAC?	Role-Based Access Control; permissions depend on roles.	Admin and Student.
36	What is ABAC?	Attribute-Based Access Control; decisions depend on attributes and context.	A student can see only their own results.
37	What is least privilege?	Give users only the permissions they need.	Student cannot delete exams.

38	What is an Identity Provider (IdP)?	A service that manages user identity and authentication.	Keycloak.
39	What is a client?	An application registered with an identity provider.	exam-frontend.
40	What is a realm?	A separate area in Keycloak for managing users, roles and clients.	exam-system.
41	What is a role?	A named group of permissions.	admin, user.

E. Data formats and password security

No.	Question	Answer in very simple words
42	What is serialization?	Converting data into a format that can be transmitted or stored.
43	What is deserialization?	Converting received data back into a usable structure.
44	What is JSON?	A lightweight data format using objects and arrays.
45	What is XML?	A data format that uses tags to organise information.
46	What is the difference between JSON and XML?	JSON uses braces and key-value pairs; XML uses opening and closing tags.
47	What is encoding?	Changing data into another representation. It is not password security.
48	What is encryption?	Protecting data so it can be recovered with a key.
49	What is hashing?	One-way conversion of data into a fixed-length result.
50	How should passwords be stored?	Use a secure salted password hash, such as Argon2 or bcrypt.
51	Why should passwords not be stored as plaintext?	Anyone who reads the database could see the real passwords.
52	Why use a salt?	It makes identical passwords produce different stored hashes.

Easy memory example: Keycloak checks the student's identity and issues a token. FastAPI checks that token and decides whether the student is allowed to perform the requested action.

Week 4 – Building the backend with FastAPI

Goal: Build APIs, validate data, connect PostgreSQL and implement CRUD.

A. FastAPI basics

No.	Question	Answer in very simple words	Example
1	What is FastAPI?	A Python framework for building APIs.	Your backend.
2	What is an endpoint?	A URL and HTTP method that perform an API operation.	GET /api/exams
3	What is a route?	The code that handles a particular API request.	A function for listing exams.
4	What is Uvicorn?	A server that runs your FastAPI application.	uvicorn app.main:app
5	What is --reload?	Automatically restarts the development server when code changes.	uvicorn app.main:app --reload
6	What is Swagger UI?	A browser interface for viewing and trying APIs.	/docs
7	What is OpenAPI?	A standard description of an API's routes, inputs and responses.	/openapi.json
8	What is a decorator?	Python syntax that attaches behaviour to a function.	@app.get("/")
9	What is a Pydantic model?	A Python class that validates data.	Validate exam title and duration.
10	What is request validation?	Checking whether received data has the correct structure and values.	Duration must be a valid number.

11	What is HTTPException?	A way to return an HTTP error from FastAPI.	Return 404 when an exam is missing.
12	What is dependency injection?	FastAPI provides required objects or services to a route.	Depends(get_db)

B. FastAPI commands in order

No.	Command	Simple meaning
13	cd .\backend	Enter backend folder.
14	python --version	Check Python.
15	python -m venv .venv	Create a Python virtual environment.
16	.\.venv\Scripts\Activate.ps1	Activate the environment.
17	python -m pip install --upgrade pip	Update pip.
18	pip install "fastapi[standard]"	Install FastAPI and its standard tools.
19	pip freeze > requirements.txt	Save installed packages.
20	uvicorn app.main:app --reload --host 127.0.0.1 --port 8000	Start FastAPI locally.
21	Ctrl + C	Stop the server.

22	cd ..	Return to project root.	

C. Database and SQL

No.	Question	Answer in very simple words	Example
23	What is SQL?	A language used to work with relational databases.	Query exam records.
24	What is PostgreSQL?	A relational database system.	Stores your examination data.
25	What is a table?	A structure that stores related records in rows and columns.	exams table.
26	What is a row?	One record in a table.	One exam.
27	What is a column?	One field in a table.	title or duration_minutes.
28	What is a primary key?	A field that uniquely identifies each row.	exam_id.
29	What is a foreign key?	A field that links one table to another.	exam_id in questions.
30	What is a parameterised query?	SQL that keeps user data separate from the query text.	Safer database input.
31	Why avoid string-formatted SQL?	It can allow SQL injection when user input is inserted unsafely.	Never concatenate user input into SQL.

32	What is an ORM?	A tool that maps database tables to program objects.	SQLAlchemy models.
33	What is SQLAlchemy?	A Python library often used for database access and ORM.	Backend database operations.
34	What is SQLite?	A lightweight file-based database.	Local practice database.
35	What is a database connection URL?	A string describing how an application connects to a database.	postgresql://...

D. Backend configuration and testing

No.	Question	Answer in very simple words
36	What is CORS?	A browser security mechanism that controls which website origins can call an API.
37	Why does your project need CORS?	React runs on port 3000 and FastAPI runs on port 8000.
38	What is an environment variable?	A value supplied to an application from outside its source code.
39	Why use environment variables?	To configure database URLs and other settings without hardcoding them.
40	What is a .env file?	A file commonly used to hold local environment settings.
41	Why should .env not be committed?	It may contain passwords and secrets.
42	What is a unit test?	A test of a small piece of code.

43	What is an API test?	A test that sends a request and checks the response.
44	What is pytest?	A Python testing framework.
45	What is 404 used for?	When a requested record does not exist.
46	What is 201 used for?	When a new record is successfully created.
47	What is 204 used for?	When an operation succeeds and returns no response body, often deletion.

E. CRUD example for your project

Operation	HTTP request	Meaning	Expected success
Create	POST /api/exams	Create an exam.	201
Read all	GET /api/exams	List exams.	200
Read one	GET /api/exams/1	Read exam 1, if implemented.	200
Replace	PUT /api/exams/1	Replace exam 1, if implemented.	200
Part update	PATCH /api/exams/1	Update some fields, if implemented.	200
Delete	DELETE /api/exams/1	Delete exam 1, if implemented.	204

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Week 5 – Virtualisation and Docker containers

Goal: Package your FastAPI application so it can run consistently on different machines.

A. Virtualisation concepts

No.	Question	Answer in very simple words	Example
1	What is virtualisation?	Creating virtual computing environments using physical hardware.	Running virtual machines.
2	What is a virtual machine (VM)?	A software-based computer with its own operating system.	Ubuntu VM.
3	What is a hypervisor?	Software that creates and manages virtual machines.	VMware.
4	What is a Type 1 hypervisor?	A hypervisor that runs directly on hardware.	VMware ESXi, Hyper-V, KVM.
5	What is a Type 2 hypervisor?	A hypervisor that runs as an application on a normal operating system.	VirtualBox.
6	What is a host OS?	The operating system of the physical host machine.	Windows on your laptop.
7	What is a guest OS?	The operating system inside a virtual machine.	Ubuntu inside VirtualBox.
8	What is a container?	An isolated environment that packages an application and its dependencies.	Your FastAPI container.

9	What is the difference between VM and container?	A VM includes a full guest OS; containers share the host kernel.	Docker containers are usually lighter.
10	What is cloud computing?	Renting computing resources over the internet.	AWS, Azure and Google Cloud.

B. Docker concepts

No.	Question	Answer in very simple words	Example
11	What is Docker?	A platform for building and running containers.	Run your backend in Docker.
12	What is Docker Engine?	The software that builds and runs Docker containers.	Docker Desktop engine.
13	What is a Dockerfile?	A recipe that describes how to build an image.	backend/Dockerfile
14	What is a Docker image?	A packaged, read-only template used to create containers.	exam-backend:1.0
15	What is a container?	A running or stopped instance created from an image.	exam-backend container.
16	What is a registry?	A place to store and share images.	Docker Hub.
17	What is the difference between image and container?	Image is the template; container is the instance created from it.	One image can create many containers.

18	What is Docker Hub?	An online registry for Docker images.	Upload your backend image.
19	What is a Docker volume?	Persistent storage managed by Docker.	Save PostgreSQL data.
20	What is a Docker network?	A network that lets containers communicate.	Backend talks to PostgreSQL.
21	What is port mapping?	Connecting a computer's port to a container's port.	-p 8000:8000.
22	What is a restart policy?	A rule for restarting a container.	unless-stopped.

C. Dockerfile questions

No.	Question	Answer in very simple words	Your backend example
23	What does FROM do?	Selects the starting image.	FROM python:3.12-slim
24	What does WORKDIR do?	Sets the working folder inside the image.	WORKDIR /app
25	What does COPY do?	Copies files into the image.	COPY app ./app
26	What does RUN do?	Runs a command while building the image.	RUN pip install ...
27	What does EXPOSE do?	Documents the port the application uses.	EXPOSE 8000
28	What does CMD do?	Defines the default command when the container starts.	Run Uvicorn.

29	Why copy requirements before app source?	Dependencies can be cached if the requirements file hasn't changed.	COPY requirements.txt . first.
30	Why use --host 0.0.0.0?	Allows the application to listen on all container network interfaces.	Makes the API reachable through Docker port mapping.
31	Why use a non-root user?	Reduces the permissions available to the application.	Run the API as a normal user.
32	What is .dockerignore?	Lists files Docker should leave out of the build context.	Ignore .venv, .git, __pycache__.

D. Docker commands

No.	Command	Simple meaning
33	docker pull python:3.12-slim	Download a Python image.
34	docker images	List images.
35	docker rmi IMAGE_NAME	Delete an image that is not in use.
36	docker build -t exam-backend:1.0 .	Build an image from the Dockerfile in the current folder.
37	docker run -d -p 8000:8000 --name exam-backend-practice exam-backend:1.0	Start a container in the background.

38	docker ps	List running containers.
39	docker ps -a	List all containers.
40	docker logs exam-backend-practice	Display container logs.
41	docker logs -f exam-backend-practice	Follow logs live.
42	docker exec -it exam-backend-practice sh	Open a shell inside the container.
43	docker stop exam-backend-practice	Stop the container.
44	docker rm exam-backend-practice	Remove the stopped container.
45	docker login	Sign in to Docker Hub.
46	docker tag exam-backend:1.0 YOUR_USERNAME/online-exam-backend:1.0	Give the image its Docker Hub name.
47	docker push YOUR_USERNAME/online-exam-backend:1.0	Upload the image to Docker Hub.

E. Important Docker questions

Question	Easy answer
In -p 9000:8000, which port does the browser use?	9000, the left side (host port).
Which port does the application use inside the container?	8000, the right side (container port).
Why does the database data survive container deletion?	Because the data is stored in a named volume, not only inside the container.
How does the backend reach the database container?	Through its Docker network and service/container name.
Why not use localhost for the database inside the backend container?	localhost means the backend container itself, not the database container.
What does docker build do?	Builds an image.
What does docker run do?	Creates and starts a container.
What does docker exec do?	Runs a command inside a running container.

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Week 6 – Docker Compose, Keycloak and complete application

Goal: Run the complete Online Examination System with one Compose command.

A. Docker Compose concepts

No.	Question	Answer in very simple words	Your project example
1	What is Docker Compose?	A tool for defining and running multiple containers together.	Runs all four services.
2	What is a Compose file?	A YAML file describing services, networks, volumes and settings.	docker-compose.yml
3	What is a service?	A definition for an application/container in Compose.	postgres, keycloak, backend, frontend.
4	What is build?	Tells Compose where to find instructions to build an image.	context: ./backend
5	What is image?	Specifies an image to use.	postgres:16-alpine
6	What is container_name?	Gives a specific name to a container.	exam-backend.
7	What is ports?	Maps host ports to container ports.	"8000:8000".
8	What is environment?	Supplies configuration values to a container.	DATABASE_URL.
9	What is volumes?	Mounts persistent storage or files into containers.	postgres_data.

10	What is networks?	Connects containers to a shared network.	exam-network.
11	What is depends_on?	Defines service startup dependencies.	Backend depends on PostgreSQL.
12	What is a health check?	A command used to check whether a service is ready.	pg_isready.
13	What is service_healthy?	Waits for a dependency's health check to pass before starting a dependent service.	Backend waits for PostgreSQL.
14	What is service_started?	Waits for the dependency container to start, not necessarily become fully ready.	Keycloak dependency.
15	What is a named volume?	Persistent storage with a Docker-managed name.	postgres_data.
16	What is a bridge network?	A private Docker network for containers to communicate.	exam-network.

B. Your four services

Service	What it does	Your settings
PostgreSQL	Stores application data.	examdb, port 5432
Keycloak	Manages login, users, clients and roles.	Realm exam-system, port 8080
FastAPI	Handles API requests and business logic.	Port 8000
React + Nginx	Displays the website to users.	Port 3000 on your computer, port 80 inside the container

C. Keycloak questions

No.	Question	Answer in very simple words
17	What is Keycloak?	An identity and access-management tool.
18	What is a realm?	A separate area for managing users, roles and clients.
19	What is your realm name?	exam-system.
20	What is a client?	An application registered in Keycloak.
21	What is your client ID?	exam-frontend.
22	What is a realm role?	A role managed inside a realm.
23	Which roles does your project use?	admin and user.
24	What is an access token?	A token a client sends to an API to prove authenticated access.
25	What is JWT?	A token format containing claims and a cryptographic signature.
26	What is the issuer (iss)?	A claim identifying who issued the token.
27	What is JWKS?	A set of public keys used to verify tokens.
28	What is token validation?	Checking the token's signature, issuer, expiry and other required claims.
29	What is realm_access.roles?	A common JWT claim location for Keycloak realm roles.

30 Why does FastAPI check the token? To confirm identity and permissions before allowing protected actions.

D. Your project's authentication flow

How student login works

1. Student opens React

localhost:3000

2. Keycloak login

Checks username and password

3. Keycloak returns an access token

Contains identity and role claims

4. React calls FastAPI

Sends the token in the Authorization header

5. FastAPI validates token and role

Allows or rejects the requested operation

6. FastAPI uses PostgreSQL

Reads or saves examination data

E. Docker Compose commands

No.	Command	Simple meaning
31	docker compose config	Validate and display Compose configuration.
32	docker compose config --services	List service names.
33	docker compose build	Build application images.
34	docker compose up	Start services and show logs in the terminal.
35	docker compose up -d	Start services in the background.
36	docker compose up -d --build	Rebuild images and start services in the background.
37	docker compose ps	Check service status.
38	docker compose logs backend	Show backend logs.
39	docker compose logs -f backend	Follow backend logs.
40	docker compose exec backend sh	Open a shell inside backend.
41	docker compose exec postgres psql -U postgres -d examdb	Open PostgreSQL.
42	docker compose down	Stop and remove containers and network, preserving named volumes.

43	docker compose down -v	Remove named volumes too; this can delete your database data.
44	docker volume ls	List Docker volumes.
45	docker network ls	List Docker networks.

Remember: Do not use docker compose down -v when you want to keep your examination records.

F. Capstone and delivery questions

No.	Question	Answer in very simple words
46	What is a capstone project?	A project that brings together the skills learned in the course.
47	What should the frontend contain?	Forms, navigation, tabs, a modal and API communication.
48	What should the backend contain?	CRUD APIs, validation, correct status codes and authentication.
49	What should the database contain?	Real application data stored persistently.
50	Why use Keycloak?	To manage authentication and roles rather than writing your own login/token system.
51	Why use .env.example?	It tells another developer which environment variables are required without exposing real secrets.
52	Why ignore .env in Git?	It may contain private credentials.

53	What should README contain?	Architecture, run commands and API endpoint list.
54	What is an architecture diagram?	A picture showing how frontend, backend, database and identity provider connect.
55	How do you prove database persistence?	Create data, run docker compose down, start again and check the data remains.
56	Why use service names instead of localhost?	Containers use Docker DNS to reach other services by their names.
57	Why use a health check?	To avoid starting dependent services before the database is ready.
58	Does the capstone require cloud deployment?	No. The guide says it runs locally using Docker Compose.